

9(a)	1 oxygen concentration decreases (with time) ; 2 oxygen concentration becomes constant when nitric oxide is added ; 3 oxygen concentration remains constant for longer with each addition ; <i>max three for explanation:</i> 4 (nitric oxide may) prevent oxygen uptake ; 5 (nitric oxide may) inhibit, pyruvate entering mitochondrion / the link reaction / Krebs cycle / ETC / oxidative phosphorylation ; 6 so oxygen cannot act as final electron acceptor ; 7 inhibits aerobic respiration ; 8 effect of nitric oxide is temporary ;	4																		
9(b)	<i>any four from:</i> <table border="1"> <thead> <tr> <th>mp</th><th>muscle</th><th>yeast</th></tr> </thead> <tbody> <tr> <td>1</td><td>lactic acid / lactate</td><td>ethanol</td></tr> <tr> <td>2</td><td>one-step process A from equation</td><td>two-step process A from equations</td></tr> <tr> <td>3</td><td>reversible</td><td>irreversible</td></tr> <tr> <td>4</td><td>no carbon dioxide produced / no decarboxylation</td><td>carbon dioxide produced / decarboxylation</td></tr> <tr> <td>5</td><td>lactate dehydrogenase</td><td>ethanol / alcohol, dehydrogenase</td></tr> </tbody> </table>	mp	muscle	yeast	1	lactic acid / lactate	ethanol	2	one-step process A from equation	two-step process A from equations	3	reversible	irreversible	4	no carbon dioxide produced / no decarboxylation	carbon dioxide produced / decarboxylation	5	lactate dehydrogenase	ethanol / alcohol, dehydrogenase	4
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Question	Answer	Marks
9(c)	any three from: 1 aerenchyma in, stems / roots, to provide oxygen to submerged parts ; 2 tall stems / fast growth of stems / greater internode growth, to hold, leaves / flowers, above water / exposed to air ; 3 roots have, ethanol / alcohol, dehydrogenase or roots are tolerant to ethanol for, ethanol fermentation / anaerobic respiration ; 4 ridges in submerged leaves to trap air ;	3